

AYESH RANJAN BHUYAN

Full Stack Software Engineer | GenAI Application Developer

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PROFESSIONAL SUMMARY

Full Stack Software Engineer with 2+ years of experience designing and building enterprise web applications, Generative AI platforms, and real-time monitoring systems. Skilled in developing Generative AI applications using Large Language Models (LLMs), LangChain, Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, and Vector Databases. Proven track record building AI-powered document validation platforms, resume screening systems, semantic search applications, intelligent workflow automation, and real-time enterprise dashboards with React.js, Node.js, Fastify, AWS, MongoDB, and PostgreSQL. Strong foundation in frontend development, backend API design, system architecture, machine learning workflows, and scalable production-grade engineering. Adept at combining Artificial Intelligence with modern full-stack development to deliver measurable business impact.

CORE COMPETENCIES

Programming Languages: Python, JavaScript, TypeScript, C++, Java, SQL, C#, C, Rust

Artificial Intelligence: LLMs, Generative AI, RAG, LangChain, LangGraph, LlamaIndex, AI Agents, Prompt Engineering, MCP, Semantic Search, Vector Databases, Embeddings, OCR, Document Intelligence, OpenAI API, Gemini API

Machine Learning: Data Preprocessing, Feature Engineering, Model Evaluation, AI Pipeline Development, Intelligent Document Processing, Resume Parsing, NLP Applications

Frontend: React.js, Next.js, Angular, Redux Toolkit, React Query, Tailwind CSS, Bootstrap, HTML5, CSS3

Backend: Node.js, Express.js, Fastify, Django, REST APIs, WebSocket, Socket.IO, JWT Authentication

Databases: PostgreSQL, MongoDB, SQL Server, MySQL, Firebase, Redis

Cloud & DevOps: AWS, Docker, Kubernetes, CI/CD, GCP

Tools & Practices: Git, GitHub, Linux, VS Code, Jira, Agile/Scrum, Postman

PROFESSIONAL EXPERIENCE

GenAI Application Developer

Jul 2025 – Present

Areca Embedded Systems Pvt. Ltd.

- Design and develop AI-powered enterprise applications using React.js, Node.js, AWS, and Fastify integrated with Large Language Models for intelligent document validation, automated approval workflows, and report generation.
- Build scalable AI workflows leveraging Prompt Engineering, Retrieval-Augmented Generation (RAG), semantic search, embeddings, and OCR-based document intelligence to automate enterprise processes.
- Develop secure backend REST APIs supporting document processing, authentication, workflow orchestration, and real-time analytics for production AI applications.
- Architect and build an enterprise-grade TCAS/KAVACH Network Monitoring System (NMS) for Indian Railways using React.js, Node.js, Fastify, SQL Server, and WebSocket communication.
- Implement real-time monitoring dashboards using UDP and WebSocket protocols, enabling continuous, low-latency monitoring of railway communication systems.
- Develop high-performance binary packet parsers in C++ to process railway telemetry and communication protocol data at scale.
- Build enterprise reporting modules including Loco Movement, Fault Monitoring, Interlocking, Parameters, Track Profile, Radio Communication, and TSRMS reports.
- Integrate AI-assisted document validation and intelligent recommendation features to streamline operational workflows and reduce manual review effort.
- Design optimized file-processing pipelines for analyzing large BIN files, implement Excel/CSV/PDF export modules, and maintain secure role-based access control (RBAC) with audit logging.
- Own end-to-end SDLC responsibilities — requirement analysis, development, debugging, testing, deployment, and production support — while participating in Agile ceremonies and peer code reviews.

Software Developer

Mar 2024 – Feb 2025

Retrack Automation Pvt. Ltd.

- Developed responsive enterprise web applications using Angular and React.js to support business-critical workflows.
- Built RESTful backend APIs using Node.js and Express.js integrated with MongoDB and MySQL databases.
- Implemented reusable frontend components that improved application maintainability and scalability.

- Owned tasks across the full Software Development Life Cycle, including requirement gathering, implementation, debugging, testing, and deployment.
- Collaborated with cross-functional Agile teams, participated in code reviews, and optimized backend API performance to resolve production issues and improve responsiveness.

PROJECTS

AI Resume Shortlist System

- Built an AI-powered recruitment platform that automates resume screening by matching candidate profiles to job descriptions using intelligent skill extraction and ranking.
- Developed React.js frontend dashboards enabling recruiters to search, filter, review, and manage shortlisted candidates through an intuitive interface.
- Implemented backend services in Node.js and Python for resume parsing, keyword extraction, semantic matching, and AI-based candidate ranking.
- Designed NLP-driven resume analysis workflows to identify relevant technical skills, experience, and education from uploaded resumes.
- Built customizable ranking algorithms that score candidates based on job-specific requirements, experience, and skill relevance.
- Implemented secure authentication, role-based access control, application status tracking, and recruiter workflow management with scalable MongoDB data models.

Technologies: Python, React.js, JavaScript, Node.js, Express.js, MongoDB, NLP, JWT Authentication, REST APIs

TCAS / KAVACH Network Monitoring System (NMS)

- Developed a large-scale Network Monitoring System for Indian Railways KAVACH (TCAS) infrastructure to monitor locomotive and stationary communication devices in real time.
- Migrated legacy Qt/C++ desktop software into a scalable React.js web platform while maintaining compatibility with existing railway communication protocols.
- Designed real-time monitoring dashboards using React.js, WebSocket, and UDP communication for continuous visualization of network health and device status.
- Built high-performance backend services using Node.js, Fastify, SQL Server, and C++ for processing railway telemetry and communication data.
- Developed binary packet parsers in C++ to decode railway communication protocols and process high-volume telemetry packets efficiently.
- Implemented enterprise reporting modules (Loco Movement, Fault Monitoring, Parameters, Interlocking, Track Profile, TSRMS, Radio Communication) with Excel/PDF/CSV export and optimized BIN file processing pipelines.
- Collaborated with cross-functional engineering teams across development, debugging, deployment, testing, and production support.

Technologies: React.js, Node.js, Fastify, SQL Server, C++, Docker, AWS, UDP, WebSocket, REST APIs

Mood Music Recommendation System

- Developed an AI-assisted web application that detects user emotions from uploaded images and generates personalized Spotify playlists based on mood analysis.
- Designed backend REST APIs using Node.js and Express.js for image processing, mood analysis, and music recommendation workflows.
- Integrated Spotify OAuth 2.0 authentication and the Spotify Web Playback SDK for secure login, playlist generation, and in-browser music streaming.
- Built a responsive Angular 17 frontend supporting image upload, mood prediction visualization, playlist browsing, and playback controls.
- Implemented efficient state management, asynchronous API communication, and ngrok-based HTTPS tunneling for secure Spotify OAuth callbacks during development.
- Optimized cross-browser compatibility and responsive UI using Angular, CSS, and SCSS for a consistent user experience across devices.

Technologies: Angular 17, TypeScript, Node.js, Express.js, Spotify SDK, OAuth 2.0, REST APIs, AI Integration

EDUCATION

B.Tech in Computer Science & Engineering

Biju Patnaik University of Technology (BPUT)

CGPA: 8.2